

Course Code : ECT 356

KQLR/RS – 24 / 1172

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

COMPUTER ARCHITECTURE

Time : 2 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Assume suitable data wherever necessary.
 - (2) All questions are compulsory.
-
1. (a) Implement the functions using PAL :
 $f_1(x, y, z) = \pi M(1, 3, 5, 7, 8, 12, 13, 14, 15)$
and
 $f_2(x, y, z) = \Sigma m(0, 1, 2, 6)$ 5(CO4)
(b) Explain concept of tags in Von Neumann architecture. 5(CO1)

 2. (a) Perform Booth Multiplication algorithm on $(-56)_{10}$ and $(-67)_{10}$. Flowchart is expected. 10(CO2)

 3. Design Control unit of Full subtractor in a Computer system using Delay Element Method and State Table Method. 10(CO5)

 4. (a) Describe DMA controller in detail. 5(CO3)
(b) Discuss Cache Write Policy. 5(CO3)

KQLR/RS-24 / 1172

Contd.

5. (a) Consider a paging system in which main memory has the capacity of three pages. The execution of a program Q requires reference to five distinct pages P_i where $i = 1, 2, 3, 4, 5$ and i is the page address. The page stream formed by executing Q is :

2 3 1 1 5 1 4 4 3 2 5 2

Assess how pages are assigned to main memory using FIFO, LRU and OPT replacement policies. 5(CO3)

- (b) Discuss the working of Vector processor. 5(CO3)
6. (a) Justify the need of Micro programmed control unit with respect to modern computer architectures. 5(CO5)
- (b) Design zero crossing detectors with respect to IC technology. 5(CO1)

